

Yongho Shin

Postdoctoral researcher

Institute of Computer Science, University of Wrocław

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RESEARCH INTERESTS

Online/approximation/learning-augmented algorithms for combinatorial optimization problems

EDUCATION

Ph.D. in Computer Science, Yonsei University, South Korea Mar. 2018 – Aug. 2024

◇ Dissertation topic: Relaxing hard requirements of online optimization via learning augmentation and limited revocability

◇ Advisor: Hyung-Chan An

B.S. in Computer Science, Yonsei University, South Korea Mar. 2012 – Feb. 2018

◇ Awarded *high honors at graduation*

EMPLOYMENT

Institute of Computer Science, University of Wrocław, Poland Nov. 2024 – present

◇ Postdoctoral researcher

◇ Advisor: Jarosław Byrka

RESEARCH PAPERS

Published Work

- [1] Changyeol Lee, Dahoon Lee, Jongseo Lee, **Yongho Shin**, and Changki Yun. Optimal learning-augmented algorithm for online bidding. *arXiv preprint arXiv:2605.07349*, 2026. **To appear in NeurIPS 2026.**
- [2] Fateme Abbasi, Martin Böhm, Jarosław Byrka, Matin Mohammadi, and **Yongho Shin**. Servicing matched client pairs with facilities. *arXiv preprint arXiv:2602.19680*, 2026. **Appeared in WAOA 2026.**
- [3] Jarosław Byrka and **Yongho Shin**. Online rounding schemes for edge cover. **Appeared in WAOA 2026.**
- [4] Fateme Abbasi, Hyung-Chan An, Jarosław Byrka, Changyeol Lee, and **Yongho Shin**. Approximating (weighted) chromatic correlation clustering via cluster LP. In *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques (APPROX/RANDOM 2026)*, 13:1-13:20, 2026.
- [5] **Yongho Shin** and Phanu Vajanopath. Parsimonious learning-augmented online metric matching. *arXiv preprint arXiv:2605.26886*, 2026. **Appeared in ICML 2026.**
- [6] Davin Choo, Billy Jin, and **Yongho Shin**. Learning-augmented online bipartite fractional matching. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 7671–7686, 2025.
- [7] Changyeol Lee, **Yongho Shin**, and Hyung-Chan An. Improved algorithms for overlapping and robust clustering of edge-colored hypergraphs: An LP-based combinatorial approach. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 70823–70855, 2025.

- [8] **Yongho Shin**, Changyeol Lee, Gukryeol Lee, and Hyung-Chan An. Improved learning-augmented algorithms for the multi-option ski rental problem via best-possible competitive analysis. *ACM Transactions on Algorithms* 22(1), 11:1–31, 2025. *Conference version in ICML 2023 [9]*.
- [9] **Yongho Shin**, Changyeol Lee, Gukryeol Lee, and Hyung-Chan An. Improved learning-augmented algorithms for the multi-option ski rental problem via best-possible competitive analysis. In *Proceedings of the 40th International Conference on Machine Learning (ICML 2023)*, PMLR 202:31539-31561, 2023.
- [10] Kangsan Kim, **Yongho Shin**, and Hyung-Chan An. Constant-factor approximation algorithms for parity-constrained facility location and k -center. *Algorithmica* 85, 1883–1911, 2023. *Conference version in ISAAC 2020 [12]*.
- [11] **Yongho Shin** and Hyung-Chan An. Making three out of two: Three-way online correlated selection. In *Proceedings of the 32nd International Symposium on Algorithms and Computation (ISAAC 2021)*, 49:1-49:17, 2021.
- [12] Kangsan Kim, **Yongho Shin**, and Hyung-Chan An. Constant-factor approximation algorithms for the parity-constrained facility location problem. In *Proceedings of the 31st International Symposium on Algorithms and Computation (ISAAC 2020)*, 21:1-21:17, 2020.

Preprints

- [13] **Yongho Shin**, Changyeol Lee, and Hyung-Chan An. On optimal consistency-robustness trade-off for learning-augmented multi-option ski rental. *arXiv preprint arXiv:2312.02547*, 2023. *Merged to [8]*.
- [14] **Yongho Shin**, Kangsan Kim, Seungmin Lee, and Hyung-Chan An. Online graph matching problem with a worst-case reassignment budget. *arXiv preprint arXiv:2003.05175*, 2020.

TEACHING

University of Wrocław, Poland

◊ Algorithms with predictions

Summer 2026

Teaching assistant, Yonsei University, South Korea

◊ CSI2103/CCO2103 Data structures

Spring 2018 – 2021, 2023, 2024

◊ CSI3108 Algorithm analysis

Fall 2018 – 2021, 2023

◊ AIC2130 Computer algorithms for AI applications

Fall 2023

◊ GEK6205 Design and analysis of optimization algorithms

Fall 2023

AWARDS

High honors at graduation, Yonsei University

Feb. 2018

ACADEMIC SERVICE

◊ Reviewer, WAOA 2026

◊ Reviewer, Discrete Optimization, 2026

◊ Reviewer, NeurIPS 2026

◊ Reviewer, MFCS 2026

◊ Reviewer, IWOCA 2026

◊ Reviewer, ICML 2026

- Gold reviewer (top 25%)

- ◇ Reviewer, Acta Informatica, 2026
- ◇ Program committee member, EC 2026
- ◇ Reviewer, STOC 2026
- ◇ Reviewer, APPROX 2025
- ◇ Reviewer, ICALP 2025
- ◇ Reviewer, Mathematical Programming, 2024
- ◇ Reviewer, WAOA 2024

TALKS AND PRESENTATIONS

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| <i>New Frontiers in Learning-Augmented Algorithms</i> , Berlin, Germany | Aug. 2026 |
| ◇ Title: Learning-augmented online bipartite fractional matching | |
| ◇ Contributed talk | |
| <i>APPROX 2026</i> , Boston, MA, USA | Aug. 2026 |
| ◇ Title: Approximating (weighted) chromatic correlation clustering via cluster LP | |
| <i>ALPS@IJCAI</i> , Bremen, Germany | Aug. 2026 |
| ◇ Title: Optimal learning-augmented algorithm for online bidding | |
| ◇ Contributed talk | |
| <i>Combinatorial Algorithms Seminar</i> , Inha University, Incheon, South Korea | July 2026 |
| ◇ Title: Primal-dual algorithms for edge-colored clustering | |
| <i>ICML 2026</i> , Seoul, South Korea | July 2026 |
| ◇ Title: Parsimonious learning-augmented online metric matching | |
| ◇ Poster presentation | |
| <i>HALG 2026</i> , Stockholm, Sweden | June 2026 |
| ◇ Title: Learning-augmented online bipartite fractional matching | |
| ◇ Contributed talk and poster presentation | |
| <i>SNU Math Seminar</i> , SNU, Seoul, South Korea | Jan. 2026 |
| ◇ Title: Online rounding for set cover under subset arrivals | |
| <i>Combinatorics Seminar</i> , Hanyang University, Seoul, South Korea | Jan. 2026 |
| ◇ Title: Primal-dual algorithms for edge-colored clustering | |
| <i>SNU CSE Seminar</i> , SNU, Seoul, South Korea | Dec. 2025 |
| ◇ Title: Learning-augmented online bipartite fractional matching | |
| <i>NeurIPS 2025</i> , San Diego, CA, USA | Dec. 2025 |
| ◇ Title: Improved algorithms for overlapping and robust clustering of edge-colored hypergraphs: An LP-based combinatorial approach | |
| ◇ Poster presentation | |
| <i>NeurIPS 2025</i> , San Diego, CA, USA | Dec. 2025 |
| ◇ Title: Learning-augmented online bipartite fractional matching | |
| ◇ Poster presentation | |
| <i>2025 Combinatorics Workshop</i> , IBS DIMAG, Daejeon, South Korea | Aug. 2025 |
| ◇ Title: Learning-augmented online bipartite fractional matching | |

◇ Contributed talk	
<i>DGIST BRL AGSTA Seminar</i> , DGIST, Daegu, South Korea	Aug. 2025
◇ Title: Online rounding for set cover under subset arrivals	
<i>Discrete Analysis Seminar</i> , Yonsei University, Seoul, South Korea	Aug. 2025
◇ Title: Online rounding for set cover under subset arrivals	
<i>Combinatorial Optimization Seminar</i> , Yonsei University, Seoul, South Korea	Aug. 2025
◇ Title: Learning-augmented online bipartite fractional matching	
<i>HALG 2025</i> , Zurich, Switzerland	June 2025
◇ Title: Learning-augmented algorithms for the multi-option ski rental problem	
◇ Contributed talk and poster presentation	
<i>COG Seminar</i> , University of Wrocław, Wrocław, Poland	May 2025
◇ Title: Learning-augmented online bipartite fractional matching	
<i>COG Seminar</i> , University of Wrocław, Wrocław, Poland	Dec. 2024
◇ Title: Online correlated selection	
<i>Discrete Analysis Seminar</i> , Yonsei University, Seoul, South Korea	June 2024
◇ Title: Three-way online correlated selection	
<i>Discrete Math Seminar</i> , IBS DIMAG, Daejeon, South Korea	May 2024
◇ Title: Three-way online correlated selection	
<i>ICML 2023</i> , Honolulu, HI, USA	July 2023
◇ Title: Improved learning-augmented algorithms for the multi-option ski rental problem via best-possible competitive analysis	
◇ Poster presentation	
<i>Theory Tea</i> , Cornell University, Ithaca, NY, USA	Dec. 2022
◇ Title: Three-way online correlated selection	
<i>ISAAC 2021</i> , Fukuoka, Japan (virtual)	Dec. 2021
◇ Title: Making three out of two: Three-way online correlated selection	
<i>AAAC 2021</i> , Tainan, Taiwan (virtual)	Oct. 2021
◇ Title: Making three out of two: Three-way online correlated selection	
<i>ISAAC 2020</i> , Hong Kong, China (virtual)	Dec. 2020
◇ Title: Constant-factor approximation algorithms for the parity-constrained facility location problem	

EXPERIENCE

Co-organizer of Yonsei CS theory student group, Yonsei University, South Korea	Jan. 2023 – Feb. 2024
◇ Initiated a reading group of TCS students in and out of Yonsei University	
◇ Organized seminar talks on various topics including mechanism design and quantum computing	
Intern, Cornell University, USA	Sept. 2022 – Dec. 2022
◇ Director: David B. Shmoys	
Undergraduate intern, Yonsei University, South Korea	Jan. 2017 – Feb. 2018

◊ Advisor: Hyung-Chan An

Undergraduate voluntary tutor, Yonsei University, South Korea

◊ CSI3108 Algorithm analysis

Fall 2016, 2017

◊ CSI2103 Data structures

Spring 2017

Web programmer, Republic of Korea Air Force, South Korea

Nov. 2013 – Aug. 2015

◊ In fulfillment of mandatory military service